

Shoaib Mohammed

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SUMMARY

Infrastructure and Systems Engineer with 10 years of production experience designing, automating, and operating large-scale on-premise cloud and virtualisation platforms — including a **323-node Kubernetes estate serving ~100,000 users** in a telecom-grade environment. Deep hands-on background across Kubernetes, OpenStack, Red Hat OpenShift, and bare-metal provisioning, with recent focus on GPU inference infrastructure and agentic AI systems. Proven track record converting manual operational processes into reproducible automation at scale. Commencing MSc in Artificial Intelligence at the University of Edinburgh (September 2026).

EXPERIENCE

Senior Member of Technical Staff — Mavenir Systems Pvt. Ltd

July 2017 – January 2025 · India | Promoted from Member of Technical Staff

- Operated and maintained a **323-node on-premise Kubernetes estate** serving **~100,000 users** in a production telecom-grade environment where public managed cloud was not an option
- Engineered automated infrastructure provisioning using **Ansible, Python, and OpenStack Heat templates**, replacing engineer-to-engineer, document-driven OS, Kubernetes, and OpenStack installation workflows; **reduced deployment time from 1.5 days to 4 hours** and eliminated copy-paste rebuild errors
- Architected and managed **OpenStack private cloud** environments on bare-metal hardware, covering compute, networking, and storage virtualisation layers
- Administered **Red Hat OpenShift** and vanilla Kubernetes clusters across on-premise and colocated infrastructure, ensuring telecom-grade availability and reliability
- Led technical evaluation of emerging cloud platform and virtualisation technologies, stress-testing candidate systems to operational limits and defining integration strategy with company products
- Collaborated cross-functionally to define platform architecture, integration patterns, and operational runbooks for cloud and HPC-adjacent workloads

Member of Technical Staff — LeadSOC

April 2025 – October 2025 · India

- Designed and built an **agentic LLM coding assistant** for FPGA design engineers using **Ollama, DeepSeek R1, and LangGraph**, engineering the orchestration and tool-invocation layer end-to-end
- Delivered internal automation tooling for HR and Finance teams, replacing manual spreadsheet-driven workflows with reproducible, programmatic pipelines
- Gained operational insight into **GPU inference economics** — model serving throughput and latency constraints — informing infrastructure sizing and deployment decisions

Software Engineer I — Hewlett-Packard Enterprise

April 2015 – July 2017 · India

- Built a **virtual machine orchestration dashboard** demonstrated at **Mobile World Congress 2016**, integrating VM provisioning, load generation, and live performance telemetry into a single operator interface
- Developed automated test frameworks for a Brocade Virtual Private Cloud, replacing manual verification cycles and accelerating release validation

RESEARCH PROJECTS

Multi-GPU Inference Scaling Study

Independent · 2026 · In Progress

- Deployed a **Qwen model across multiple GPUs on RunPod** to measure inference throughput and latency response to GPU scaling
- Investigating GPU serving economics as the binding operational constraint for production LLM systems — directly applicable to HPC/AI infrastructure sizing

Transformer Architecture Implementation — LLaMA Conversion

Independent · 2025–2026

- Implemented a transformer language model **from first principles**, progressively replacing components to arrive at the LLaMA architecture
- Covered **rotary positional embeddings (RoPE)**, **RMSNorm**, **SwiGLU activations**, and **grouped-query attention (GQA)** — building architecture-level understanding of modern LLM internals

EDUCATION

MSc Artificial Intelligence — University of Edinburgh

September 2026 – 2027 · Commencing

B.Tech — JNTU Anantapur

2014

SKILLS

Cloud & Virtualisation: Kubernetes (bare-metal, production), Red Hat OpenShift, OpenStack (compute/network/storage), VMware concepts, private cloud architecture **HPC & AI Infrastructure:** GPU inference deployment (RunPod, Ollama), multi-GPU workload management, LLM serving, inference optimisation **Automation & IaC:** Ansible, Terraform, OpenStack Heat Templates, Python, Bash **Agentic & ML Systems:** LangGraph, Ollama, Hugging Face, PyTorch, DeepSeek R1 **Platforms & Environments:** Linux (RHEL, Ubuntu), bare-metal provisioning, telecom-grade on-premise and colocated infrastructure **Languages:** Python, Bash, YAML